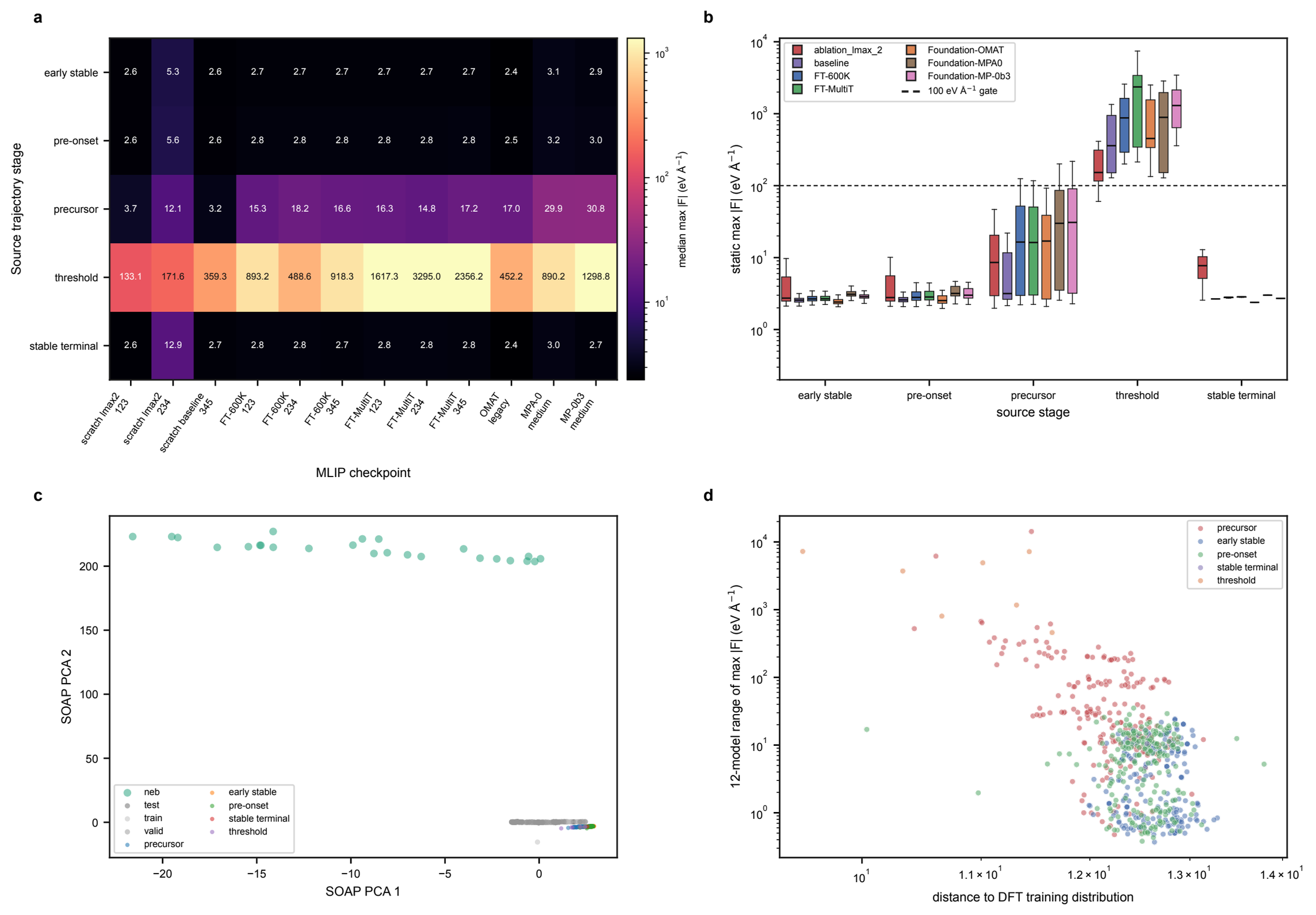

a
Source trajectory stage
early stable
pre-onset
precursor
threshold
stable terminal
2.6 5.3 2.6 2.7 2.7 2.7 2.7 2.7 2.7 2.4 3.1 2.9
2.6 5.6 2.6 2.8 2.8 2.8 2.8 2.8 2.8 2.5 3.2 3.0
3.7 12.1 3.2 15.3 18.2 16.6 16.3 14.8 17.2 17.0 29.9 30.8
133.1 171.6 359.3 893.2 488.6 918.3 1617.3 3295.0 2356.2 452.2 890.2 1298.8
2.6 12.9 2.7 2.8 2.8 2.7 2.8 2.8 2.8 2.4 3.0 2.7
median max |F| (eV Å⁻¹)
scratch lmax2 123
scratch lmax2 234
scratch baseline 345
FT-600K 123
FT-600K 234
FT-600K 345
FT-MultiT 123
FT-MultiT 234
FT-MultiT 345
OMAT legacy
MPA-0 medium
MP-0b3 medium
MLIP checkpoint
b
static max |F| (eV Å⁻¹)
ablation_lmax_2
baseline
FT-600K
FT-MultiT
Foundation-OMAT
Foundation-MPA0
Foundation-MP-0b3
100 eV Å⁻¹ gate
early stable
pre-onset
precursor
threshold
stable terminal
source stage
c
SOAP PCA 2
SOAP PCA 1
neb
test
train
valid
precursor
early stable
pre-onset
stable terminal
threshold
d
12-model range of max |F| (eV Å⁻¹)
precursor
early stable
pre-onset
stable terminal
threshold
distance to DFT training distribution